

Software Engineering Lab 1:

Remote Patient Vitals Alert & Monitoring App

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1. Problem Overview

- Post-operative and chronic care patients require continuous monitoring of vital signs (SpO₂, Heart Rate, Blood Pressure).
- The system ingests telemetry data, detects threshold breaches, and escalates alerts to caregivers in real time and ensures timely intervention to reduce risk of unattended emergencies.

2. Identify Functional Requirements

SI no.	Requirement	Priority	Acceptance Criteria	Rationale
FR001	Continuously ingest patient vitals (SpO ₂ , HR, BP) from wearable sensors.	High	Data stream is received every 5 seconds without interruption.	Enables real-time monitoring of patient health.
FR002	Compare incoming vitals against configured clinical thresholds.	High	If HR > 140 BPM or SpO ₂ < 90%, system flags anomaly within 2 seconds.	Detects emergencies quickly.
FR003	Generate alerts and notify assigned caregiver when thresholds are breached.	High	Caregiver receives push notification and SMS within 5 seconds of anomaly detection.	Ensures timely intervention.
FR004	Maintain patient history logs for vitals and alerts.	Medium	System stores timestamped vitals and alerts retrievable for at least 30 days.	Supports medical review and auditing.
FR005	Allow caregiver to acknowledge and respond to alerts.	High	Once caregiver confirms, alert status changes to "Acknowledged" and response time is logged.	Tracks accountability and response efficiency.

3. Identify Non Functional Requirements

ID	Requirement	Priority	Acceptance Criteria	Rationale
NFR001	System uptime must be $\geq 99.99\%$ for telemetry ingestion.	High	Benchmark tests confirm uptime under simulated peak load.	Ensures reliability in critical healthcare scenarios.
NFR002	Patient data must be encrypted in transit and at rest.	High	All telemetry streams use TLS; logs stored with AES-256 encryption.	Protects sensitive health information.

4. Actors

- Patient (Remote): Provides vitals via wearable sensors.
- Caregiver (On-Call): Receives alerts, acknowledges, and responds.
- System Gateway: Ingests telemetry, applies thresholds, generates alerts.

5. Use Cases

ID	Use Case
UC-01	Ingest Patient Vitals
UC-02	Detect Threshold Breach
UC-03	Generate Alert
UC-04	Notify Caregiver
UC-05	Acknowledge Alert
UC-06	Review Patient History

6. Use Case Flow

Main Success Scenario:

1. The patient's wearable streams vitals to the system.
2. System ingests and evaluates against thresholds.
3. An anomaly is detected.

4. Alert is generated.
5. Caregiver receives notification.
6. Caregiver acknowledges alert.
7. System logs response time and stores vitals history.

Alternate Flows:

1. No Anomaly Detected:
 - Vitals remain within thresholds.
 - No alert generated.
 - System continues monitoring.
2. Caregiver Unavailable:
 - Primary caregiver does not acknowledge within 2 minutes.
 - System escalates alert to secondary caregiver.

7. Diagram

